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Dump truck

Abstract

A dump truck is provided, which includes a vehicle frame and a vehicle body hinged on the vehicle frame, a front wheel installation shaft and a rear wheel installation shaft respectively arranged on front and rear ends of the vehicle frame, and wheels installed on the front wheel installation shaft and the rear wheel installation shaft, a connecting plate fixedly installed below a rear end of the vehicle body, and a body rotation shaft fixed below the connecting plate, the connecting plate and the vehicle frame are rotatably connected through the body rotation shaft, and the body rotation shaft is arranged directly above the rear wheel installation shaft.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) The present application is a 35 U.S.C. § 371 National Phase conversion of International (PCT) Patent Application No. PCT/CN2021/110199 filed Aug. 3, 2021, which claims foreign priority of

TECHNICAL FIELD

(2) The present disclosure relates to the technical filed of dump truck, and in particular to a new type of dump truck.

BACKGROUND

(3) Dump truck is a kind of vehicle for short-distance transportation of materials with a tippable body. Due to its low cost, simple maintenance, convenient operation and wide application, it is suitable for short-distance transportation of sand, earth, coal, ore or other bulk materials. Transport is a very practical transport device in the process of daily handling of objects. Most of the existing dump trucks use a rotating shaft to connect the body to the frame. Since the body is usually made of a plastic one-piece material, the connection point with the rotating shaft is usually not resistant to wear and tear, resulting in serious wear and tear of the connection point after using the tipping function for a long time, thus increasing the replacement frequency of the body. Moreover, most of the existing dump trucks are designed to be pushed by hand, so it is not convenient to use a trailer hook for towing when heavy goods are loaded in the truck.

(4) Therefore, the existing technology still needs to be further developed and improved.

SUMMARY

(5) In view of various deficiencies of the prior art, in order to solve the above problems, a new type of dump truck is proposed. The present disclosure provides the following technical solutions:

(6) A new type of dump truck, comprises a vehicle frame and a vehicle body hinged on the vehicle frame, a front wheel installation shaft and a rear wheel installation shaft respectively arranged on front and rear ends of the vehicle frame, and wheels installed on the front wheel installation shaft and the rear wheel installation shaft, a connecting plate fixedly installed below the rear end of the vehicle body, and a body rotation shaft fixed below the connecting plate, the connecting plate and the vehicle frame are rotatably connected through the body rotation shaft, and the body rotation shaft is arranged directly above the rear wheel installation shaft.

(7) Further, a connecting frame is fixed below the connecting plate, the body rotation shaft is arranged on the connecting frame, and the connecting plate is connected with the body rotation shaft through the connecting frame.

(8) Further, the connecting frame is a V-shaped connecting frame, comprising a first straight rod vertically arranged, an second straight rod inclined arranged, and a connecting curved rod connecting both ends of the first straight rod and the second straight rod, the first straight rod, the connecting curved rod and the second straight rod are jointly enclosed to form the V-shaped connecting frame, and the body rotation shaft is arranged at the connection between the first straight rod and the connecting curved rod.

(9) Further, the connecting plate is two corner plates, and the two corner plates are respectively arranged above the rear wheel installation shaft.

(10) Further, the rear end of the vehicle frame extends downward to form two rear wheel installation frames, and the body rotation shaft and the rear wheel installation shaft are arranged up and down and mounted on the rear wheel installation frames.

(11) Further, the rear wheel installation shaft is a single long rotating shaft, and the two ends of the single long rotating shaft are respectively rotatably connected to the two rear wheel installation frames.

(12) Further, A front wheel installation frame is fixedly connected below the front end of the vehicle frame, the front wheel installation frame is a U-shaped installation frame, and two front wheel installation shafts are respectively connected to two sides of the U-shaped installation frame.

(13) Further, A connecting rod is fixedly connected to the front wheel installation frame, and a reversible handle is hinged on the end of the connecting rod.

(14) Further, A connecting rod is fixedly connected to the front wheel installation frame, and a

retractable handle is hinged on the end of the connecting rod.

(15) Further, a handle is arranged on the front end of the vehicle body for easily turning the vehicle body, a snap-fit buckle for connecting the vehicle frame is arranged below the handle, and a snap-fit plate is arranged on the vehicle frame corresponding to the snap-fit buckle, and the snap-fit plate is fixedly connected to the vehicle frame, the snap-fit buckle and snap-fit plate are detachably snap-fitted

Beneficial Effects

(16) 1. By setting the body rotation shaft just above the rear wheel installation shaft and below the vehicle frame plane, that is, the body rotation shaft is located between the vehicle frame and the rear wheel installation shaft, reducing the rotation center of gravity when the vehicle body is turned over, through simple structural changes to achieve the purpose of labor saving, to ensure that the rotation center is above the front wheel rotation axis, to avoid the rotation center of gravity is too low to affect the complete overturning of the vehicle body.

(17) 2. By setting the connecting plate, the connection area between the hinged structure and the vehicle body is increased, thereby reducing the wear of the vehicle body during long-term overturning and prolonging the service life of the dump truck; by setting the connecting frame, it is used to support the connecting plate and disperse the gravity of the vehicle body on the body rotation shaft, which prolongs the service life of the body rotation shaft.

(18) 3. By setting the V-shaped connecting frame with stable structure, and setting the body rotation shaft at the connection between the first straight rod and the connecting curved rod, the center of gravity of the vehicle body is shifted inside the triangle when the vehicle body is turned to the vertical state. In this way, the pressure of the vehicle body on the body rotation shaft is reduced, the service life of the body rotation shaft is prolonged, and at the same time, the center of gravity of the goods in the vehicle body is between the front and rear wheels during the overturning process, so as to avoid the suspension of the front wheel caused by the center of gravity exceeding the rear wheel.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a structural schematic diagram of the new type of dump truck according to an embodiment 1 of the present disclosure.

(2) FIG. 2 is a structural schematic diagram of the new type of dump truck after removing the vehicle body and wheels according to the embodiment 1 of the present disclosure.

(3) FIG. 3 is a left view structural schematic diagram of the new type of dump truck after removing the vehicle body and wheels according to the embodiment 1 of the present disclosure.

(4) FIG. 4 is top view structural schematic diagram of the new type of dump truck after removing the vehicle body and wheels according to the embodiment 1 of the present disclosure.

(5) FIG. 5 is rear view structural schematic diagram of the new type of dump truck after removing the vehicle body and wheels according to the embodiment 1 of the present disclosure.

(6) FIG. 6 is a structural schematic diagram of the extension rod of the reversible handle according to the embodiment 1 of the present disclosure.

(7) FIG. 7. is a structural schematic diagram of the retractable handle according to a embodiment 2 of the present disclosure.

(8) In the figures: **100**, extension rod; **110**, first fixing hole; **120**, second fixing hole; **130**, third fixing hole; **140**, connecting piece; **200**, hand pull ring; **210**, cross bar; **220**, connecting arm; **221**, first installation hole; **222**, second installation hole; **300**, vehicle frame; **310**, rear wheel installation frame; **311**, first straight rod; **312**, second straight rod; **313**, connecting curved rod; **314**, connecting plate; **320**, front wheel installation frame; **330**, snap-fit buckle; **400**, connecting rod; **500**, vehicle

body; **510**, body rotation shaft; **600**, wheel; **610**, rear wheel installation shaft; **620**, front wheel installation shaft; **700**, handle; **710**, snap-fit plate; **800**, positioning installation hole; **900**, limit rod.

DETAILED DESCRIPTION

(9) In order to make those skilled in the art better understand the technical solutions of the present disclosure, the technical solutions of the present disclosure will be described clearly and completely below with reference to the accompanying drawings of the present disclosure. Other similar embodiments obtained by technical personnel without creative work shall fall within the protection scope of the present application. In addition, the directional terms mentioned in the following embodiments, such as “up”, “down”, “left”, “right”, etc. are only the directions of the drawings, so the directional terms used are used to illustrate rather than limit the present disclosure.

Embodiment 1

(10) As shown in FIGS. **1-6**, a new type of dump truck, includes a vehicle frame **300** and a vehicle body **500** hinged on the vehicle frame **300**, a front wheel installation shaft **620** and a rear wheel installation shaft **610** respectively arranged on front and rear ends of the vehicle frame **300**, and wheels **600** installed on the front wheel installation shaft **620** and the rear wheel installation shaft **610**, a connecting plate **314** fixedly installed below the rear end of the vehicle body **500**, and a body rotation shaft **510** fixed below the connecting plate **314**, the connecting plate **314** and the vehicle frame **300** are rotatably connected through the body rotation shaft **510**, and the body rotation shaft **510** is arranged directly above the rear wheel installation shaft **610**. By setting the body rotation shaft **510** just above the rear wheel installation shaft **610** and below the vehicle frame **300** plane, that is, the body rotation shaft **510** is located between the vehicle frame **300** and the rear wheel installation shaft **610**, reducing the rotation center of gravity when the vehicle body **500** is turned over, through simple structural changes to achieve the purpose of labor saving, to ensure that the rotation center is above the front wheel rotation axis, to avoid the rotation center of gravity is too low to affect the complete overturning of the vehicle body **500**.

(11) Further, a connecting frame is fixed below the connecting plate **314**, the body rotation shaft **510** is arranged on the connecting frame, and the connecting plate **314** is connected with the body rotation shaft **510** through the connecting frame. Since the vehicle body **500** is usually made of plastic and other easily-wearable materials, by setting the connecting plate **314**, the connection area between the hinged structure and the vehicle body **500** is increased, thereby reducing the wear of the vehicle body **500** during long-term overturning and prolonging the service life of the dump truck; by setting the connecting frame, it is used to support the connecting plate **314** and disperse the gravity of the vehicle body **500** on the body rotation shaft **510**, which prolongs the service life of the body rotation shaft **510**.

(12) Further, the connecting frame is a V-shaped connecting frame, comprising a first straight rod **311** vertically arranged, an second straight rod **312** inclined arranged, and a connecting curved rod **313** connecting both ends of the first straight rod **311** and the second straight rod **312**, the first straight rod **311**, the connecting curved rod **313** and the second straight rod **312** are jointly enclosed to form the V-shaped connecting frame, and the body rotation shaft **510** is arranged at the connection between the first straight rod **311** and the connecting curved rod **313**. By setting the V-shaped connecting frame with stable structure, and setting the body rotation shaft **510** at the connection between the first straight rod **311** and the connecting curved rod **313**, the center of gravity of the vehicle body is shifted inside the triangle when the vehicle body **500** is turned to the vertical state. In this way, the pressure of the vehicle body **500** on the body rotation shaft **510** is reduced, the service life of the body rotation shaft **510** is prolonged, and at the same time, the center of gravity of the goods in the vehicle body **500** is between the front and rear wheels during the overturning process, so as to avoid the suspension of the front wheel caused by the center of gravity exceeding the rear wheel.

(13) Further, the connecting plate **314** is two corner plates, and the two corner plates are respectively arranged above the rear wheel installation shaft. The corner plate has a simple

structure, reduces the overall weight of the vehicle body, is easy to obtain materials, and has a low manufacturing cost.

(14) Further, the rear end of the vehicle frame **300** extends downward to form two rear wheel installation frames **310**, and the body rotation shaft **510** and the rear wheel installation shaft **610** are arranged up and down and mounted on the rear wheel installation frames **310**. The rear wheel installation frame **310** is integrally designed with the vehicle frame **300** to ensure the structural strength of the rear wheel installation frame **310**, and there is no need to add additional components for fixing the rear wheel installation frame **310**, thereby reducing the cost of raw materials.

(15) Further, the rear wheel installation shaft **610** is a single long rotating shaft, and the two ends of the single long rotating shaft are respectively rotatably connected to the two rear wheel installation frames **310**. In addition to fixing the rear wheel, the long rotating shaft can also be connected with two rear wheel installation frames **310** to share the weight of the vehicle body **500** on the top vehicle frame **300**.

(16) Further, a front wheel installation frame **320** is fixedly connected below the front end of the vehicle frame **300**, the front wheel installation frame **320** is a U-shaped installation frame, and two front wheel installation shafts **620** are respectively connected to two sides of the U-shaped installation frame. The bottom of the U-shaped installation frame is flush with the bottom of the rear wheel installation frame **310**.

(17) Further, a connecting rod **400** is fixedly connected to the front wheel installation frame **320**, and a reversible handle is hinged on the end of the connecting rod **400**. The connecting rod **400** is fixedly connected to the side of the U-shaped installation frame, and the reversible handle includes an extension rod **100** hinged on the connecting rod **400** and a hand pull ring **200** connected to the extension rod **100**. The end of the hand pull ring **200** is reversibly connected with the end of the extension rod **100**, and the end of the extension rod **100** is arranged with a connecting piece **140** for connecting a trailer. The extension rod **100** is arranged with a fixing hole, and an installation hole corresponding to the hand pull ring **200**, and the extension rod **100** and the hand pull ring **200** are rotatably connected through a pin shaft passing through the installation hole and the fixing hole. By hinged connecting the hand pull ring **200** with the extension rod **100**, the hand pull ring **200** can be rotated at the end of the extension rod **100** as needed, When the hand pull ring **200** needs to be manually pulled by hand, the hand pull ring **200** is turned outward, and when the vehicle needs to be hooked, the hand pull ring **200** is turned inward, thereby exposing the connecting piece **140**, so that the connecting piece **140** is close to the vehicle to complete the quick conversion process.

Further, the hand pull ring **200** is a U-shaped ring, and the openings at both ends of the U-shaped ring are hinged to both sides of the end of the extension rod **100** respectively. The reversible connection is realized by two pairs of fixing holes arranged up and down along the direction of the extension rod **100**. The openings at both ends of the U-shaped ring are respectively provided with two pairs of installation holes corresponding to the two pairs of fixing holes, and the first pin shaft passes through the first installation hole **221** and the first fixing hole **110** below in turn, and then the U-shaped ring is hinged with the extension rod **100**. Then, the second detachable pin shaft is passed through the second installation hole **222** and the second fixing hole **120** above in turn, and then the U-shaped ring is fixedly connected to the extension rod **100**. The fixed connection between the U-shaped ring and the extension rod **100** is realized through the joint action of the first pin shaft and the second pin shaft. When it is necessary to pull the hand pull ring **200**, just pull it directly. When it is necessary to hook the vehicle for use, after pulling out the detachable second pin, turn the U-shaped ring to make the connecting piece **140** located inside the U-shaped ring. It is exposed to facilitate the hooking of the vehicle. Preferably, the extension rod **100** is provided with three pairs of fixing holes, that is, the extension rod **100** is also provided with a third fixing hole **130**, and the third fixing hole **130** is fixed below the first fixing hole **110** along the direction of the extension rod **100**, after pulling out the second pin shaft and rotating the U-shape around the first pin shaft until the second installation hole **222** and the third fixing hole **130** coincide, then insert the second

pin shaft, to realize the fixed storage of the hand pull ring **200**. Further, the hand pull ring **200** includes a cross bar **210** for easy grasping and a connecting arm **220** for hinged connecting the cross bar **210** to the extension bar **100**. There are two connecting arms **220**, which are respectively fixedly connected to both side ends of the cross bar **210**. The cross bar **210** and the connecting arm **220** are jointly enclosed to form a U-shaped ring. Further, the cross bar **210** is a cylindrical cross bar **210**, and the interior of the cylindrical cross bar **210** is hollow. The cylindrical crossbar **210** is suitable for ergonomic design, easy to grasp, and has a hollow interior design, which reduces the weight of the crossbar **210**, reduces the material used for the handle, and reduces the production cost of the enterprise without affecting the structural strength. Further, a reinforcing rib is provided on the outer side of the connecting arm **220**. By setting the reinforcing ribs, the strength of the handle can be effectively improved, the amount of materials used can be reduced, and the production cost of the enterprise can be reduced. Further, the connecting arm **220** is a V-shaped connecting arm **220**, and the included angle of the V-shaped connecting arm **220** is an obtuse angle. The V-shaped connecting arm **220** prevents the cross bar **210** from interfering with the extension bar **100** after being turned over, which affects the fixation of the second pin shaft and the third fixing hole **130**. The connecting arm **220** and the cross bar **210** are jointly enclosed to form a teardrop-shaped hand pull ring **200** structure. The teardrop-shaped structure is not only beautiful, but also can withstand a large pulling force, and can effectively support both pushing force and pulling force. Further, the extension rod **100** is a hollow square tube. The square tube is convenient to increase the contact area between the extension rod **100** and the handle ring **200**, provides greater support for the handle ring **200**, and at the same time facilitates the handle ring **200** to be effectively fixed after being turned over. Further, the end of the extension rod **100** extends upward to form a connecting piece **140**. The connecting piece **140** is arranged parallel to the installation plane of the hand pull ring **200**. The connecting piece **140** and the extension rod **100** are integrally designed to increase the structural strength of the connecting piece **140** and facilitate the transmission of force between the connecting piece **140** and the extension rod **100**. Further, the extension rod **100** is a V-shaped extension rod **100**. The included angle of the V-shaped extension rod **100** is an obtuse angle, and the V-shaped extension rod **100** is set in accordance with the mechanical principle, which is convenient to push or pull the dump truck while keeping the wheels **600** on the ground, and at the same time, it can ensure that the dump truck can be pushed or pulled when the handle is tilted forward. May be close to the vehicle body **500** for easy storage.

(18) Further, a handle **700** is arranged on the front end of the vehicle body **500** for easily turning the vehicle body **500**, a snap-fit buckle **330** for connecting the vehicle frame **300** is arranged below the handle **700**, and a snap-fit plate **710** is arranged on the vehicle frame **300** corresponding to the snap-fit buckle **330**, and the snap-fit plate **710** is fixedly connected to the vehicle frame **300**, the snap-fit buckle **330** and snap-fit plate **710** are detachably snap-fitted.

Embodiment 2

(19) The structure of the dump truck of this embodiment is basically the same as that of the embodiment 1, the only difference is that, as shown in FIG. 7, a connecting rod **400** is fixedly connected to the front wheel installation frame **320**, a retractable handle is hinged at the end of the connecting rod **400**. The retractable handle includes an extension rod **100** hinged to the end of the connecting rod **400** and a hand pull ring **200** connected to the extension rod **100**. The connecting rod **400** is horizontally fixed on the side of the front end of the vehicle frame **300** or the side of the front wheel frame. The other end of the connecting rod **400** is hinged with one end of the extension rod **100**. The extension rod **100** is a V-shaped rod, which is suitable for the push and pull of people of different heights, and is convenient at the same time. For storage, the other end of the extension rod **100** is detachably and fixedly connected with a hand pull ring **200**, which is convenient for grasping and improves the comfort of pushing and pulling person. A positioning hole is provided under the hand pull ring **200**, a positioning installation hole **800** is disposed on the extension rod **100**, and the third pin shaft passes through the positioning installation hole **800** and the positioning

hole to detachably fix the hand pull ring **200** on the extension rod **100**. Further, two sets of positioning installation holes **800** are provided on the extension rod **100** at intervals, and the distance between the two sets of positioning installation holes **800** is not less than the height of the hand pull ring **200**. Each set of positioning installation holes **800** includes two positioning installation holes **800** opposite to both sides of the extension rod **100**. When the hand pull ring **200** is fixed in the positioning installation holes **800** above through the third pin, the hand-pull ring **200** is located at the end of the extension rod **100** and higher than the extension rod **100**, which is convenient for manual push and pull. When the hand pull ring **200** is fixed in the positioning and installation hole **800** below through the third pin, the hand pull ring **200** is located in the extension inside the rod **100**, and the positioning installation holes **800** above it are higher than the highest point of the hand pull ring **200**, which is convenient for hooking and pulling the trailer. Therefore, two sets of positioning installation holes **800** and a detachable fixed hand pull ring **200** improves the scope of application of the push-pull method of the dump truck. Further, the bottom of the hand pull ring **200** is fixedly connected with a limit rod **900**, a limit hole is provided in the corresponding extension rod **100**, and the limit rod **900** extends into the limit hole so that the hand pull ring **200** is slidably connected to the extension rod **100**. By arranging the limit rod **900** and the limit hole which are slidably connected, it is convenient to quickly change the position of the hand pull ring **200**, and the installation efficiency is improved.

(20) The present disclosure has been described in detail above. The above descriptions are only preferred embodiments of the present disclosure, and should not limit the scope of implementation of the present disclosure. It still falls within the scope of the present disclosure.

Claims

1. A dump truck, comprising: a vehicle frame and a vehicle body hinged on the vehicle frame, a front wheel installation shaft and a rear wheel installation shaft respectively arranged on front and rear ends of the vehicle frame, and wheels installed on the front wheel installation shaft and the rear wheel installation shaft, a connecting plate fixedly installed below the rear end of the vehicle body, and a body rotation shaft fixed below the connecting plate, the connecting plate and the vehicle frame are rotatably connected through the body rotation shaft, and the body rotation shaft is arranged directly above the rear wheel installation shaft; wherein a connecting frame is fixed below the connecting plate, the body rotation shaft is arranged on the connecting frame, and the connecting plate is connected with the body rotation shaft through the connecting frame.
2. The dump truck according to claim 1, wherein the connecting frame is a V-shaped connecting frame, comprising a first straight rod vertically arranged, a second straight rod inclined arranged, and a connecting curved rod connecting both ends of the first straight rod and the second straight rod, the first straight rod, the connecting curved rod and the second straight rod are jointly enclosed to form the V-shaped connecting frame, and the body rotation shaft is arranged at a connection between the first straight rod and the connecting curved rod.
3. The dump truck according to claim 1, wherein the connecting plate is two corner plates, and the two corner plates are respectively arranged above the rear wheel installation shaft.
4. The dump truck according to claim 1, wherein a rear end of the vehicle frame extends downward to form two rear wheel installation frames, and the body rotation shaft and the rear wheel installation shaft are vertically arranged and mounted on the two rear wheel installation frames.
5. The dump truck according to claim 4, wherein the rear wheel installation shaft is a single long rotating shaft, and the two ends of the single long rotating shaft are respectively rotatably connected to the two rear wheel installation frames.
6. The dump truck according to claim 1, wherein a front wheel installation frame is fixedly connected below the front end of the vehicle frame, the front wheel installation frame is a U-shaped installation frame, and two front wheel installation shafts are respectively connected to two sides of

the U-shaped installation frame.

7. The dump truck according to claim 6, wherein a connecting rod is fixedly connected to the front wheel installation frame, and a reversible handle is hinged on the end of the connecting rod.

8. The dump truck according to claim 6, wherein a connecting rod is fixedly connected to the front wheel installation frame, and a retractable handle is hinged on the end of the connecting rod.

9. The dump truck according to claim 1, wherein a handle is arranged on the front end of the vehicle body for easily turning the vehicle body, a snap-fit buckle for connecting the vehicle frame is arranged below the handle, and a snap-fit plate is arranged on the vehicle frame corresponding to the snap-fit buckle, and the snap-fit plate is fixedly connected to the vehicle frame, the snap-fit buckle and snap-fit plate are detachably snap-fitted.
